

Maria Andrea Malonzo
TN35

Code:

```
#include <GL/glut.h>
#include <cstdlib>
#include <string>

const int WINDOW_WIDTH = 2167;
const int WINDOW_HEIGHT = 672;

bool dropdownOpen = false;
std::string selectedTerm = "1-20262027";

void setColor(float r, float g, float b) {
    glColor3f(r, g, b);
}

void drawRect(float x, float y, float width, float height,
              float r, float g, float b) {
    setColor(r, g, b);

    glBegin(GL_QUADS);
    glVertex2f(x, y);
    glVertex2f(x + width, y);
    glVertex2f(x + width, y + height);
    glVertex2f(x, y + height);
    glEnd();
}

void drawOutline(float x, float y, float width, float height,
                 float r, float g, float b) {
    setColor(r, g, b);

    glBegin(GL_LINE_LOOP);
    glVertex2f(x, y);
    glVertex2f(x + width, y);
    glVertex2f(x + width, y + height);
    glVertex2f(x, y + height);
    glEnd();
}
```

```

}

void drawText(float x, float y, const char* text,
             void* font = GLUT_BITMAP_HELVETICA_12,
             float r = 0.08f,
             float g = 0.15f,
             float b = 0.22f) {
    setColor(r, g, b);
    glRasterPos2f(x, y);

    for (int i = 0; text[i] != '\0'; i++) {
        glutBitmapCharacter(font, text[i]);
    }
}

void drawCenteredText(float x, float y, float width,
                     const char* text,
                     void* font = GLUT_BITMAP_HELVETICA_12,
                     float r = 0.08f,
                     float g = 0.15f,
                     float b = 0.22f) {
    int textWidth = glutBitmapLength(
        font,
        reinterpret_cast<const unsigned char*>(text)
    );

    drawText(
        x + (width - textWidth) / 2.0f,
        y,
        text,
        font,
        r,
        g,
        b
    );
}

void drawDropdownArrow(float x, float y) {
    setColor(0.72f, 0.72f, 0.72f);

```

```

    glBegin(GL_TRIANGLES);
    glVertex2f(x, y + 7);
    glVertex2f(x + 10, y + 7);
    glVertex2f(x + 5, y);
    glEnd();
}

void drawHeader() {
    drawText(
        20,
        WINDOW_HEIGHT - 54,
        "Schedule and Assessment",
        GLUT_BITMAP_HELVETICA_18,
        0.05f,
        0.10f,
        0.16f
    );

    drawText(
        WINDOW_WIDTH - 240,
        WINDOW_HEIGHT - 45,
        "Schedule and Assessment",
        GLUT_BITMAP_HELVETICA_12,
        0.05f,
        0.10f,
        0.16f
    );
}

void drawControls() {
    const float labelX = 25;
    const float controlY = WINDOW_HEIGHT - 145;

    drawText(
        labelX,
        controlY + 10,
        "Term - School Year :"
    );

    const float dropdownX = 201;

```

```
const float dropdownY = controlY - 2;  
const float dropdownW = 155;  
const float dropdownH = 41;
```

```
drawRect(  
    dropdownX,  
    dropdownY,  
    dropdownW,  
    dropdownH,  
    1.0f,  
    1.0f,  
    1.0f  
);
```

```
drawOutline(  
    dropdownX,  
    dropdownY,  
    dropdownW,  
    dropdownH,  
    0.55f,  
    0.55f,  
    0.55f  
);
```

```
drawText(  
    dropdownX + 9,  
    dropdownY + 14,  
    selectedTerm.c_str(),  
    GLUT_BITMAP_HELVETICA_12,  
    0.65f,  
    0.65f,  
    0.65f  
);
```

```
drawDropdownArrow(  
    dropdownX + dropdownW - 16,  
    dropdownY + 17  
);
```

```
drawRect(  

```

```
382,  
dropdownY - 4,  
88,  
45,  
0.10f,  
0.63f,  
0.25f  
);  
  
drawText(  
    397,  
    dropdownY + 12,  
    "Submit",  
    GLUT_BITMAP_HELVETICA_12,  
    1.0f,  
    1.0f,  
    1.0f  
);  
  
if (dropdownOpen) {  
    drawRect(  
        dropdownX,  
        dropdownY - 35,  
        dropdownW,  
        35,  
        1.0f,  
        1.0f,  
        1.0f  
    );  
  
    drawOutline(  
        dropdownX,  
        dropdownY - 35,  
        dropdownW,  
        35,  
        0.55f,  
        0.55f,  
        0.55f  
    );  
}
```

```

        drawText(
            dropdownX + 9,
            dropdownY - 22,
            "1-20262027",
            GLUT_BITMAP_HELVETICA_12,
            0.20f,
            0.20f,
            0.20f
        );
    }
}

void drawStudentInformation() {
    const float leftX = 26;
    const float rightX = 1028;
    const float startY = 421;
    const float lineHeight = 22;

    drawText(
        leftX,
        startY,
        "Enrollment Status : ENROLLED"
    );

    drawText(
        leftX,
        startY - lineHeight,
        "Student # : 202310284"
    );

    drawText(
        leftX,
        startY - lineHeight * 2,
        "Name : MALONZO, MARIA ANDREA VILLAREAL"
    );

    drawText(
        leftX,
        startY - lineHeight * 3,
        "Phone : +6396922810204    Zip Code : 1800"
    );
}

```

```

);

drawText(
    leftX,
    startY - lineHeight * 4,
    "Classification :"
);

drawText(
    rightX,
    startY - lineHeight,
    "College : COMPUTER STUDIES    Program : BSCSE"
);

drawText(
    rightX,
    startY - lineHeight * 2,
    "Year Level : 2        1st term, SY 26-27"
);

drawText(
    rightX,
    startY - lineHeight * 3,
    "Address : 13 FLAMINGO STREET, NEW MARIKINA SUBDIVISION SAN ROQUE"
);
}

void displayCourses() {
    const float tableX = 25;
    const float tableY = 45;
    const float rowH = 29;
    const float headerH = 31;

    const float colTitle = tableX + 148;
    const float colSection = tableX + 1035;
    const float colUnits = tableX + 1155;
    const float colDays = tableX + 1245;
    const float colTime = tableX + 1320;
    const float colRoom = tableX + 1810;

```

```

const float rowStart = tableY + headerH;

// CS0011

float y = rowStart + 10;

drawCenteredText(tableX + 10, y, 100, "CS0011",
                 GLUT_BITMAP_HELVETICA_10);
drawText(colTitle, y, "MOBILE PROGRAMMING",
         GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colSection - 20, y, 75, "TN35",
                 GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colUnits - 10, y, 45, "3",
                 GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colDays - 20, y, 70, "M / W",
                 GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colTime - 45, y, 390,
                 "11:00:00-12:50:00 / 11:00:00-12:50:00",
                 GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colRoom - 35, y, 180, "F608 / F1209",
                 GLUT_BITMAP_HELVETICA_10);

// CS0016
y = rowStart + rowH * 6 + 10;

drawCenteredText(tableX + 10, y, 100, "CS0016",
                 GLUT_BITMAP_HELVETICA_10);
drawText(colTitle, y, "NETWORK AND COMMUNICATIONS 2A",
         GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colSection - 20, y, 75, "TS31",
                 GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colUnits - 10, y, 45, "3",
                 GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colDays - 20, y, 70, "T / TH",
                 GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colTime - 45, y, 390,
                 "17:00:00-18:50:00 / 17:00:00-18:50:00",
                 GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colRoom - 35, y, 180, "ONLINE / F1101",
                 GLUT_BITMAP_HELVETICA_10);

```

```

// CS0019
y = rowStart + rowH * 5 + 10;

drawCenteredText(tableX + 10, y, 100, "CS0019",
                  GLUT_BITMAP_HELVETICA_10);
drawText(colTitle, y, "MODELING AND SIMULATION",
          GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colSection - 20, y, 75, "TN35",
                  GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colUnits - 10, y, 45, "3",
                  GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colDays - 20, y, 70, "F / T",
                  GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colTime - 45, y, 390,
                  "11:00:00-12:50:00 / 11:00:00-12:50:00",
                  GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colRoom - 35, y, 180, "ONLINE / ONLINE",
                  GLUT_BITMAP_HELVETICA_10);

// CS0025
y = rowStart + rowH * 4 + 10;

drawCenteredText(tableX + 10, y, 100, "CS0025",
                  GLUT_BITMAP_HELVETICA_10);
drawText(colTitle, y, "SOFTWARE ENGINEERING 1",
          GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colSection - 20, y, 75, "TN35",
                  GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colUnits - 10, y, 45, "3",
                  GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colDays - 20, y, 70, "F / TH",
                  GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colTime - 45, y, 390,
                  "15:00:00-16:50:00 / 15:00:00-16:50:00",
                  GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colRoom - 35, y, 180, "ONLINE / E609",
                  GLUT_BITMAP_HELVETICA_10);

// CS0045

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```

y = rowStart + rowH * 3 + 10;

drawCenteredText(tableX + 10, y, 100, "CS0045",
    GLUT_BITMAP_HELVETICA_10);
drawText(colTitle, y,
    "CS ELECTIVE - COMPUTER GRAPHICS AND VISUAL COMPUTING",
    GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colSection - 20, y, 75, "TN35",
    GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colUnits - 10, y, 45, "3",
    GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colDays - 20, y, 70, "M / TH",
    GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colTime - 45, y, 390,
    "13:00:00-14:50:00 / 13:00:00-14:50:00",
    GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colRoom - 35, y, 180, "E601 / E610",
    GLUT_BITMAP_HELVETICA_10);

// GED0019
y = rowStart + rowH * 2 + 10;

drawCenteredText(tableX + 10, y, 100, "GED0019",
    GLUT_BITMAP_HELVETICA_10);
drawText(colTitle, y, "UNDERSTANDING THE SELF",
    GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colSection - 20, y, 75, "TX21",
    GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colUnits - 10, y, 45, "3",
    GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colDays - 20, y, 70, "F / T",
    GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colTime - 45, y, 390,
    "07:00:00-8:50:00 / 07:00:00-8:50:00",
    GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colRoom - 35, y, 180, "ONLINE / ONLINE",
    GLUT_BITMAP_HELVETICA_10);

// GED0075
y = rowStart + rowH + 10;

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drawCenteredText(tableX + 10, y, 100, "GED0075",
                  GLUT_BITMAP_HELVETICA_10);
drawText(colTitle, y, "LINEAR ALGEBRA",
          GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colSection - 20, y, 75, "TS21",
                  GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colUnits - 10, y, 45, "3",
                  GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colDays - 20, y, 70, "F / T",
                  GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colTime - 45, y, 390,
                  "13:00:00-14:50:00 / 13:00:00-14:50:00",
                  GLUT_BITMAP_HELVETICA_10);
drawCenteredText(colRoom - 35, y, 180, "ONLINE / ONLINE",
                  GLUT_BITMAP_HELVETICA_10);
}

void drawTable() {
    const float tableX = 25;
    const float tableY = 45;
    const float tableW = WINDOW_WIDTH - 50;
    const float rowH = 29;
    const float headerH = 31;

    // Header background
    drawRect(
        tableX,
        tableY + rowH * 8,
        tableW,
        headerH,
        0.88f,
        0.88f,
        0.87f
    );

    // Alternating row backgrounds
    for (int row = 0; row < 7; row += 2) {
        drawRect(
            tableX,

```

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        tableY + rowH + row * rowH,
        tableW,
        rowH,
        0.93f,
        0.93f,
        0.92f
    );
}

// Table border
drawOutline(
    tableX,
    tableY,
    tableW,
    headerH + rowH * 8,
    0.55f,
    0.55f,
    0.55f
);

// Column headings
drawText(tableX + 8, tableY + rowH * 8 + 11, "Courses");
drawText(tableX + 148, tableY + rowH * 8 + 11, "Title");
drawText(tableX + 1035, tableY + rowH * 8 + 11, "Section");
drawText(tableX + 1155, tableY + rowH * 8 + 11, "Units");
drawText(tableX + 1245, tableY + rowH * 8 + 11, "Days");
drawText(tableX + 1320, tableY + rowH * 8 + 11, "Time");
drawText(tableX + 1810, tableY + rowH * 8 + 11, "Room");

displayCourses();

// Total row
drawRect(
    tableX,
    tableY,
    tableW,
    rowH,
    0.93f,
    0.93f,
    0.92f

```

```

);

drawText(
    tableX + 8,
    tableY + 10,
    "TOTAL UNITS 21",
    GLUT_BITMAP_HELVETICA_10,
    0.20f,
    0.27f,
    0.32f
);
}

void display() {
    glClear(GL_COLOR_BUFFER_BIT);

    drawRect(
        0,
        0,
        WINDOW_WIDTH,
        WINDOW_HEIGHT,
        0.96f,
        0.97f,
        0.98f
    );

    drawHeader();
    drawControls();
    drawStudentInformation();
    drawTable();

    glutSwapBuffers();
}

void reshape(int width, int height) {
    glViewport(0, 0, width, height);

    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();

```

```

gluOrtho2D(
    0,
    WINDOW_WIDTH,
    0,
    WINDOW_HEIGHT
);

glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
}

void mouse(int button, int state, int x, int y) {
    if (button != GLUT_LEFT_BUTTON ||
        state != GLUT_DOWN) {
        return;
    }

    float glY = WINDOW_HEIGHT - y;

    const float dropdownX = 201;
    const float dropdownY = WINDOW_HEIGHT - 147;
    const float dropdownW = 155;
    const float dropdownH = 41;

    // Open or close dropdown
    if (x >= dropdownX &&
        x <= dropdownX + dropdownW &&
        glY >= dropdownY &&
        glY <= dropdownY + dropdownH) {
        dropdownOpen = !dropdownOpen;
        glutPostRedisplay();
        return;
    }

    // Select dropdown option
    if (dropdownOpen &&
        x >= dropdownX &&
        x <= dropdownX + dropdownW &&
        glY >= dropdownY - 35 &&
        glY <= dropdownY) {

```

```

        selectedTerm = "1-20262027";
        dropdownOpen = false;
        glutPostRedisplay();
        return;
    }

    // Submit button
    if (x >= 382 &&
        x <= 470 &&
        glY >= dropdownY - 4 &&
        glY <= dropdownY + 41) {
        dropdownOpen = false;
        glutPostRedisplay();
    }
}

void keyboard(unsigned char key, int, int) {
    if (key == 27) {
        std::exit(0);
    }
}

int main(int argc, char** argv) {
    glutInit(&argc, argv);

    glutInitDisplayMode(
        GLUT_DOUBLE | GLUT_RGB
    );

    glutInitWindowSize(
        WINDOW_WIDTH,
        WINDOW_HEIGHT
    );

    glutInitWindowPosition(80, 80);

    glutCreateWindow(
        "Schedule and Assessment"
    );
}

```

Screenshot:

Schedule and Assessment		Schedule and Assessment
Term - School Year:	1-20262027	Submit
Enrollment Status : ENROLLED		
Student # : 202310284		
Name : MALONZO, MARIA ANDREA VILLAREAL		
Phone : +6396922810204 Zip Code : 1800		
Classification :		
College : COMPUTER STUDIES Program : BSCSE		
Year Level : 2 1st term, SY 26-27		
Address : 13 FLAMINGO STREET, NEW MARIKINA SUBDIVISION SAN ROQUE		

Courses	Title	Section	Units	Days	Time	Room
C90016	NETWORK AND COMMUNICATIONS 2A	T831	3	T / TH	17:00:00-18:50:00 / 17:00:00-18:50:00	ONLINE / F1101
C80019	MODELING AND SIMULATION	TN35	3	F / T	11:00:00-12:50:00 / 11:00:00-12:50:00	ONLINE / ONLINE
C90025	SOFTWARE ENGINEERING I	TN35	3	F / TH	15:00:00-16:50:00 / 15:00:00-16:50:00	ONLINE / E609
C80045	CS ELECTIVE - COMPUTER GRAPHICS AND VISUAL COMPUTING	TN35	3	M / TH	13:00:00-14:50:00 / 13:00:00-14:50:00	E601 / E610
GED0019	UNDERSTANDING THE SELF	TX21	3	F / T	07:00:00-8:50:00 / 07:00:00-8:50:00	ONLINE / ONLINE
GED0075	LINEAR ALGEBRA	TS21	3	F / T	13:00:00-14:50:00 / 13:00:00-14:50:00	ONLINE / ONLINE
C90011	MOBILE PROGRAMMING	TN35	3	M / W	11:00:00-12:50:00 / 11:00:00-12:50:00	F608 / F1209
TOTAL UNITS 21						